

# KIMANI, Josiah Kiarie

AI, Data & Software Engineer | [josiah.k.kimani@gmail.com](mailto:josiah.k.kimani@gmail.com) | +254700261018 | [linkedIn](#) | [github](#)

## PROFESSIONAL SUMMARY

I am a recent Google DeepMind MSc graduate with Cum Laude (Distinction) in Mathematical Sciences (specialising in Artificial Intelligence (AI) for Science) from the African Institute for Mathematical Sciences (AIMS) – South Africa, accredited by Stellenbosch University. I hold a first-class honours BSc in Mathematical Sciences with IT and have a deep passion for leveraging AI and data-driven solutions to address global challenges.

My expertise spans machine learning, deep learning (Jax, PyTorch, TensorFlow) in climate science, computer vision, GenAI, LLMs and Reinforcement Learning, alongside experience in software engineering; backend (Java, PHP), frontend (ExtJs, ReactJs), database management (MySQL, PostgreSQL) and cloud computing(Google Cloud Platform).

I am particularly interested in developing and applying state-of-the-art AI models for climate prediction and weather forecasting/nowcasting to improve accuracy and quantify uncertainty in extreme weather events attributed to climate change.

## KEY SOFT SKILLS & PERSONAL ATTRIBUTES

- Responsive leader and team player.
- Self-motivated and innovative.
- Analytical problem solving.
- Observant and keen on details.
- Excellent oral and written communication skills.

## EDUCATION

| Duration  | Certification                                                         | Institution                                  |
|-----------|-----------------------------------------------------------------------|----------------------------------------------|
| 2024-2025 | MSc. Mathematical Sciences - AI for Science (Cum Laude)               | Stellenbosch University & AIMS-South Africa. |
| 2016-2021 | BSc. Mathematical Sciences, With I.T. (1 <sup>ST</sup> Class Honours) | Maseno University, Kenya.                    |
| 2012-2015 | High School Diploma (Grade: B +)                                      | Kenswed Secondary School.                    |

## WORK EXPERIENCE

**Feb 2026 - Present: Research Associate**, AfriClimate AI

*Duties and Responsibilities:*

- Supporting the development of an AI-based seasonal weather forecasting model for South Africa.
- Collecting and preprocessing climate datasets
- Designing and experimenting AI models
- Running AI models evaluation and benchmarks
- Contributing to reports and publications

**Sept 2025 - Jan 2026: Program Assistant**, Leadership Development Program (LDP) at Emerging Leaders Foundation(ELF)-Africa

*Duties and Responsibilities:*

- Implementing and coordinating projects and activities (iLEAD Fellowship, NOW US! Awards & St

Andrews Turi Leadership Award).

- Designing and developing concept notes and proposals for proposed projects and initiatives within the LDP department.
- Developing relevant programmatic content/curricula for the LDP department and other departments.

**Sept 2024 - July 2025: Google DeepMind Scholar** at AIMS-South Africa.

*Duties and Responsibilities:*

- Undertaking a one-year fully funded Master's in Mathematical Sciences (AI for Science).
- Supporting community engagement activities through the AIMS-South Africa House of Science.
- Coordinating the AIMS-South Africa Running club to promote mental wellness, fitness and community building.

**Sept 2023 - Aug 2024: Program Assistant**, Leadership Development Program (LDP) at Emerging Leaders Foundation(ELF)-Africa

*Duties and Responsibilities:*

- Implementing and coordinating projects and activities (iLEAD Fellowship, NOW US! Awards & St Andrews Turi Leadership Award).
- Designing and developing concept notes and proposals for proposed projects and initiatives within the LDP department.
- Developing relevant programmatic content/curricula for the LDP department and other departments.

**Nov 2020 - Aug 2023: Full-stack Software Developer** at Smart People Africa Limited Company.

*Duties and Responsibilities:*

- Developing, implementing and upgrading the PickPay Enterprise Resource Planning (E.R.P) Project for 35 water utility companies in Kenya.
- Organising and training users of the Enterprise Resource Planning Water utility system.
- Establishing and evaluating Standard Operating Procedures (SOP) for better teamwork, order and quality control spanning technical and Customer Support departments.

## PROJECTS

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- **Master's Thesis:** "PiggyCast: Improving Weather Prediction Accuracy through a Stacking-Based Ensemble AI Approach." ([GitHub Repo](#))
- **Applied Machine Learning At Scale - The MovieLens Recommender System:** "Matrix factorisation Recommendation Algorithm on the 32 million MovieLens Dataset."([GitHub Repo](#))
- **Deep Learning for Ecology Soundscapes Classifier:** "Modelling Soundscapes (Bio-phoney, Geo-phoney and Anthro-phoney) of the Environment through Deep Learning."([GitHub Repo](#))

## PROFESSIONAL LICENSES AND CERTIFICATIONS

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| Issued Date | Certification                                                   | Institution                                                |
|-------------|-----------------------------------------------------------------|------------------------------------------------------------|
| April 2025: | Africa Scientifique Program                                     | AIMS-South Africa & African Gong.                          |
| Feb 2023:   | Certificate of Completion of iLEAD Fellowship Cohort 9          | ELF-Africa.                                                |
| July 2021   | Microsoft Dynamics 365 Business Central Supply Chain Essentials | Use Dynamics.                                              |
| June 2021   | Programming   HTML5 with JavaScript and CSS3   PHP   WordPress  | Kenswed Technical and Vocational Training College (KTVTC). |
| June 2021   | Graphic Design with Adobe Illustrator                           | KTVTC.                                                     |
| Mar 2021    | Microsoft Dynamics 365 Business Central Finance Essentials      | Use Dynamics.                                              |
| Dec 2020    | Introduction to Java Programming: Java Fundamental Concepts     | Coursera.                                                  |
| Dec 2020    | Programming Foundations with JavaScript, HTML and CSS           | Coursera.                                                  |
| Jan 2017    | Certificate in Peace Ambassador & Peer Counselling              | Maseno University.                                         |

## ACHIEVEMENTS AND RESPONSIBILITIES

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Jan 2025 - July 2025: **Volunteer** – AIMS-South Africa House of Science.

Oct 2024 - July 2025: **President** – AIMS-South Africa Running Club.

2023 - 2024: **Financial Secretary** – Santa Rosa De Lima Choir and Ngong Cathedral Brass Band.

2022 - Present: **Organizing Secretary** – iLEAD fellowship program cohort nine (9).

2019 - Present: **President** – Kenswed Secondary School Alumni Association.

2017 - 2020: **Peace Ambassador and Peer Counsellor**– Maseno University.

2018 - 2019: **Chairperson** – Catholic Students' Association, Maseno University.

2018 - 2019: **Vice-Chairperson** – St. Anne's Chaplaincy Council, Maseno University.

## REFEREES

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To be provided on request.